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Export System

▼ Relevant source files

`admin/setup/dboperations/dailyCronJobs.php``admin/utilities/cleanupFoldersDBs.php``admin/utilities/notifyDatabaseArchive.php``admin/utilities/purgeInactiveDBs.php``admin/verification/checkRecURL.php``export/dbbackup/buildArchivePackagesCMD.php``export/dbbackup/exportMyDataPopup.php``export/xml/flathml.php``hclient/widgets/database/dbAction.js``hclient/widgets/database/dbVerify.html``hserv/controller/databaseController.php``hserv/structure/export/getDBStructureAsSQL.php``hserv/structure/export/getDBStructureAsXML.php`

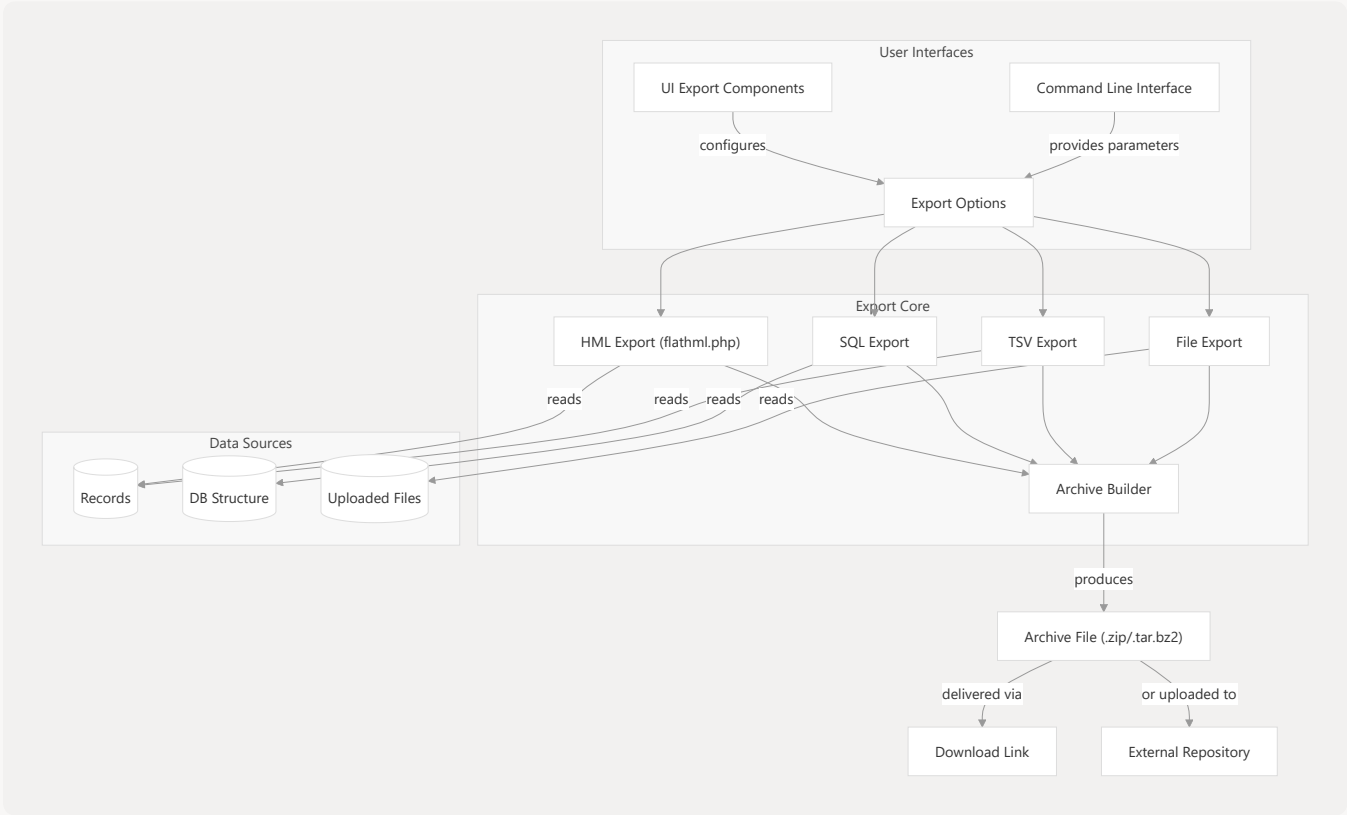
The Export System in Heurist provides functionality for exporting data and database structures in various formats, creating comprehensive backups, and archiving databases. It supports both user-initiated exports through the UI and automated exports through [command-line tools](#) and cron jobs. For related functionality that validates data before export, see [Data Management](#).

Export Formats and Capabilities

Heurist supports multiple export formats, each serving different purposes:

Format	Description	Primary Use Cases
HML (Heurist Markup Language)	Custom XML format preserving record relationships	Long-term archiving, data interchange
SQL	Database structure and data as SQL statements	Database migration, backup
TSV (Tab-Separated Values)	Tabular data format	Spreadsheet import, data analysis
ZIP/BZip Archive	Combined package of multiple formats and files	Comprehensive backup, database transfer

Export System Architecture

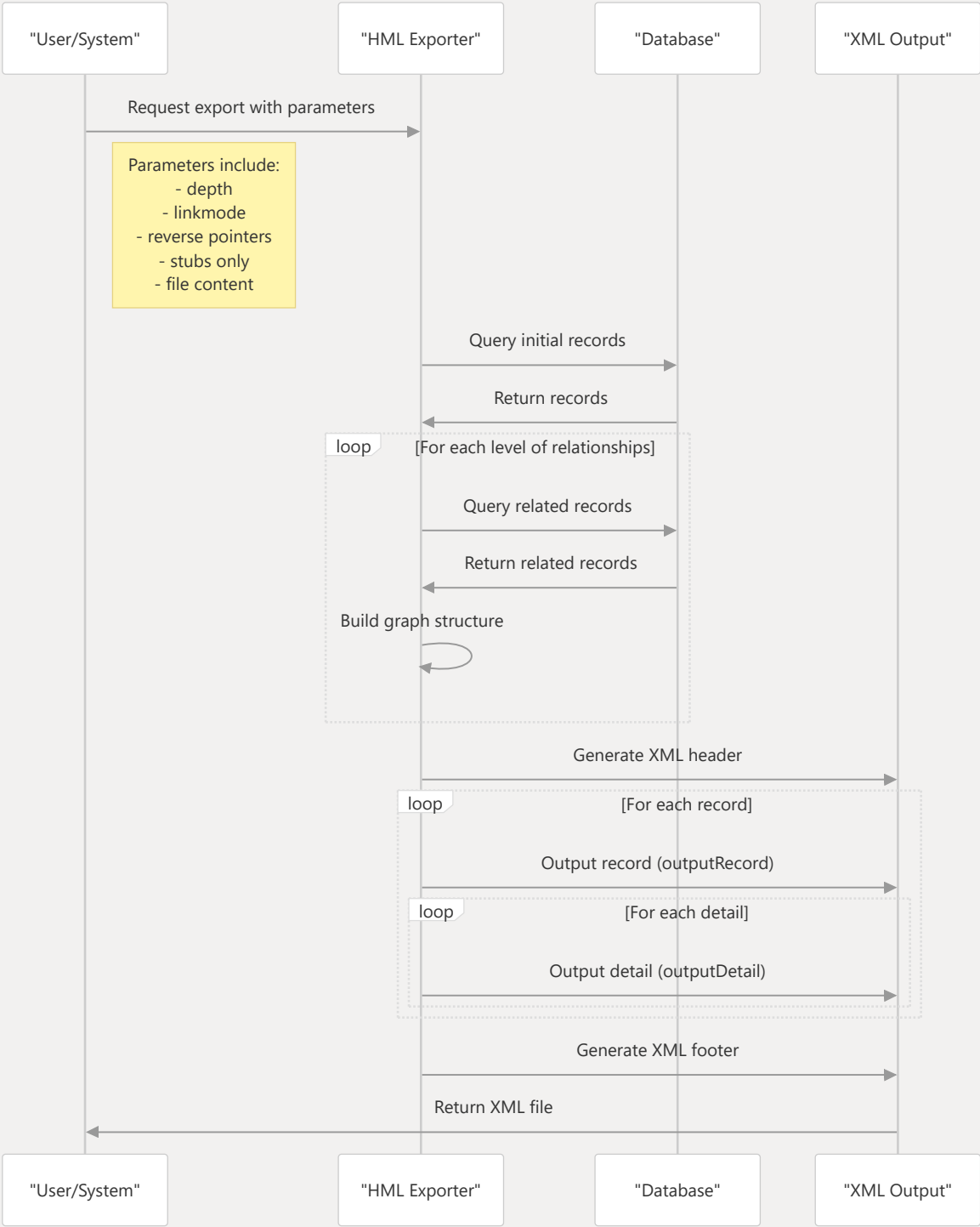


Sources: `export/xml/flathml.php` `export/dbbackup/exportMyDataPopup.php`
`export/dbbackup/buildArchivePackagesCMD.php`

HML (Heurist Markup Language) Export

The HML export is implemented in `flathml.php` and produces a custom XML format that preserves the relationships between records while avoiding redundancy.

HML Export Process



Sources: `export/xml/flathml.php` 512-584 `export/xml/flathml.php` 596-663
`export/xml/flathml.php` 677-772

Key HML Export Parameters

Parameter	CLI Flag	Description
REVERSE	rev	Include reverse pointer fields (default: yes)
EXPAND_REV_PTR	revexpand	Expand reverse pointers (default: yes)

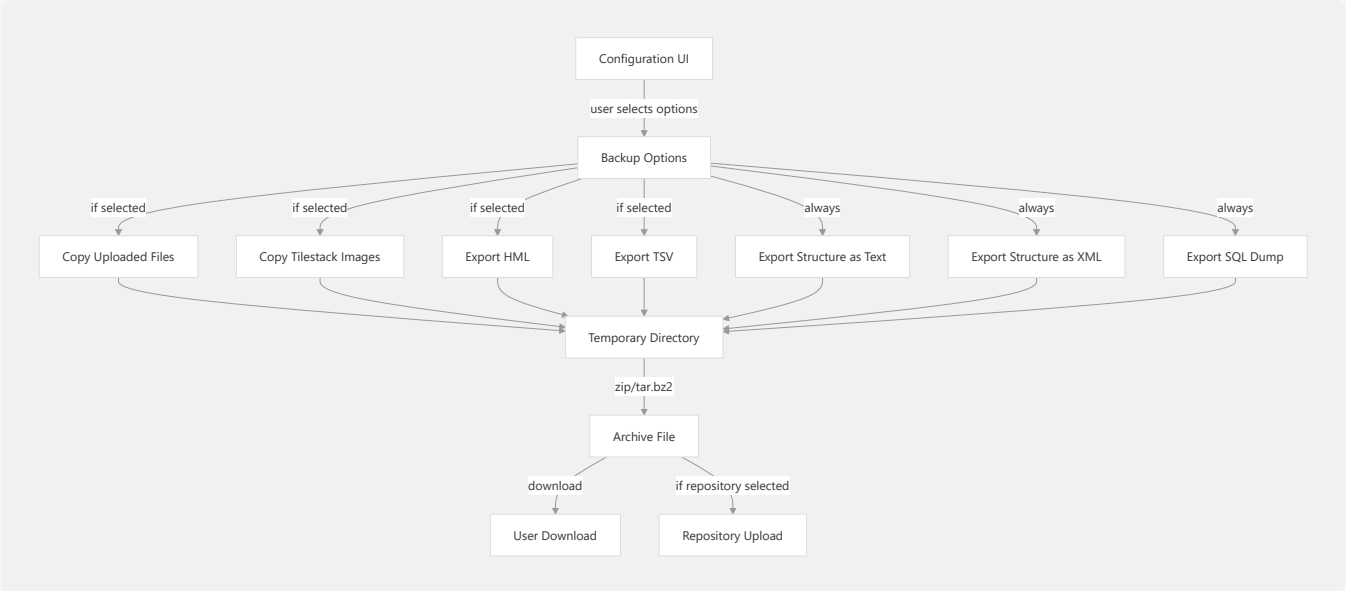
Parameter	CLI Flag	Description
<code>NO_RELATIONSHIPS</code>	N/A	Don't output relationships
<code>WOOT</code>	<code>woot</code>	Don't output text content (default: no)
<code>USEXINCLUDE</code>	<code>hinclude</code>	Use xinclude format for related records
<code>OUTPUT_STUBS</code>	<code>stub</code>	Output only record header fields
<code>INCLUDE_FILE_CONTENT</code>	<code>fc</code>	Expand XML file content
<code>depth</code>	<code>depth</code>	Control how many levels of related records are included
<code>linkmode</code>	<code>linkmode</code>	Control how links are handled (none, direct, direct_links, all)

Sources: `export/xml/flathml.php` 341-401

Database Backup

The database backup functionality in Heurist creates comprehensive archives of databases, including data, structure, files, and documentation. It's primarily implemented in `exportMyDataPopup.php`.

Database Backup Process



Sources: `export/dbbackup/exportMyDataPopup.php` 310-314

`export/dbbackup/exportMyDataPopup.php` 443-478 `export/dbbackup/exportMyDataPopup.php` 493-645

Backup Options

The backup process supports the following options:

1. **Include attached files** - Copies uploaded files into the archive
2. **Include tiled map images** - Copies tilestack images (typically very large)
3. **Include HML** - Adds a fully self-documenting HML file
4. **Include TSV** - Adds a folder containing the database dump in TSV format
5. **Include documentation** - Adds system documentation for long-term archiving
6. **Archive format** - ZIP (default) or BZip
7. **Repository upload** - Option to upload the archive to an external repository (like Nakala)

Sources: `export/dbbackup/exportMyDataPopup.php` 329-401

Command-line Export Tools

Heurist provides command-line tools for creating exports and backups, primarily for automation and batch processing.

Build Archive Packages

The `buildArchivePackagesCMD.php` tool creates archive packages for one or more databases:

```
sudo php -f buildArchivePackagesCMD.php -- -db=database_1,database_2
sudo php -f buildArchivePackagesCMD.php -- -db=all -nofiles -nodoc
```

Command-line options:

- `-db=<database_list>` - Specify which database(s) to archive (comma-separated, or "all")
- `-nofiles` - Skip including uploaded files
- `-nodoc` - Skip including documentation
- `-nosql` - Skip including SQL export
- `-nohml` - Skip including HML export
- `-notsv` - Skip including TSV export

Sources: `export/dbbackup/buildArchivePackagesCMD.php` 38-90

`export/dbbackup/buildArchivePackagesCMD.php` 160-251

Purge Inactive Databases

The `purgeInactiveDBs.php` tool archives and optionally deletes inactive databases based on specific criteria:

```
sudo php -f purgeInactiveDBs.php -- -purge
```

Criteria for archiving:

- 3 months inactive with 10 records or less
- 6 months inactive with 50 records or less
- 1 year inactive with 200 records or less

Sources: `admin/utilities/purgeInactiveDBs.php` 79-103

`admin/utilities/purgeInactiveDBs.php` 301-324

Structure Export

Heurist provides functionality to export the database structure in both XML and SQL formats.

XML Structure Export

The XML structure export (`getDBStructureAsXML.php`) generates an XML representation of the database structure, including record types, detail types, terms, and other schema elements.

Key features:

- XML representation of the database structure
- Option to include user and group information
- Option to export a subset of the structure by record type, detail type, or term ID

Sources: `hserv/structure/export/getDBStructureAsXML.php` 80-146

`hserv/structure/export/getDBStructureAsXML.php` 202-224

SQL Structure Export

The SQL structure export (`getDBStructureAsSQL.php`) generates SQL INSERT statements for recreating the database structure.

Key features:

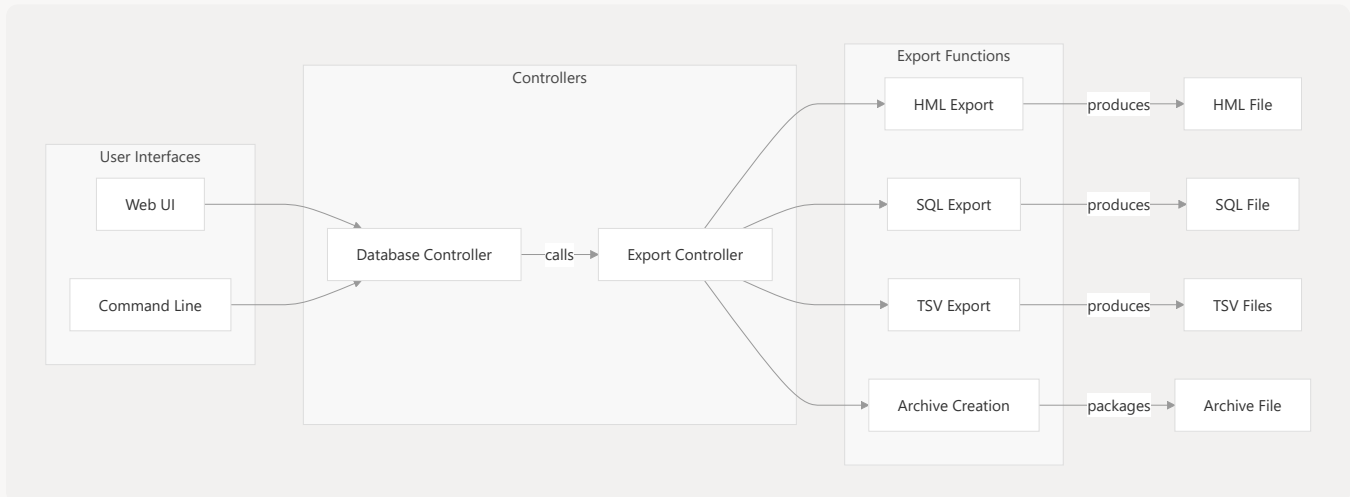
- SQL INSERT statements for each definition table
- Option to include user and group information
- Option to filter for approved definitions only

Sources: `hserv/structure/export/getDBStructureAsSQL.php` 66-106`hserv/structure/export/getDBStructureAsSQL.php` 170-202

Integration with Database Management

The Export System integrates with the database management functionality in Heurist. The

`databaseController.php` file handles various database operations, including exports and backups.

Sources: `hserv/controller/databaseController.php` 240-249`export/dbbackup/exportMyDataPopup.php` 55-146

Automated Exports and Archiving

Heurist includes functionality for automated exports and archiving through cron jobs and scheduled processes.

Daily Cron Jobs

The `dailyCronJobs.php` script runs various daily tasks, which can include URL validation related to exports:

```
sudo php -f dailyCronJobs.php -- url
```

Sources: `admin/setup/dboperations/dailyCronJobs.php` 234-299

Database Archiving Notifications

The `notifyDatabaseArchive.php` script sends notifications to database owners about creating archives of their databases if they've been modified within the last month:

```
sudo php -f notifyDatabaseArchive.php -- -db=database_1,database_2
```

Sources: `admin/utilities/notifyDatabaseArchive.php` 126-185

Export System Data Flow

The following diagram illustrates the data flow during a complete database backup process:

